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Roll No. 24/AFI/14

FIRST SEMESTER

M.Tech. (AI)

MID SEMESTER EXAMINATION

Sept, 2024

AI507/RIRA505 NATURAL LANGUAGE PROCESSING

Time: 1:30 Hours

Max. Marks: 20

Note: Answer **ALL** questions.
Assume suitable missing data, if any.
CO# is course outcome(s) related to the question.

1. Given the following corpus of sentences: "Dogs love to play in the park," "Cats prefer staying at home," and "Dogs and cats can live together peacefully," use a Bi-gram Language Model to calculate the probability of the sentence: "Cats love the park." If necessary, apply a smoothing technique, such as Laplace smoothing. [5] [CO3]
2. You are building a document retrieval system and decide to use TF-IDF to evaluate the importance of terms within your corpus. Consider the following three documents:
Document 1 (D1): Data science is fascinating.
Document 2 (D2): Machine learning and data analysis are key in data science."
Document 3 (D3): "Artificial intelligence overlaps with machine learning."
 - [a] List all unique terms in the corpus after converting them to lowercase and removing punctuation and determine the term frequency for each unique term in D2.
 - [b] Calculate the inverse document frequency for the term "data." Compute the TF-IDF score for "data" in Document 2 (D2).
 - [c] Finally, explain what this TF-IDF score indicates about the significance of the term "data" in the context of your document collection. [2+2+1] [CO3, CO1]
3. You are tasked with developing a text classification system to categorize customer feedback into "Positive" or "Negative" sentiments using Logistic Regression. Explain how Logistic Regression determines the decision boundary between the two sentiment classes. Illustrate how the model uses the learned coefficients to make classification decisions. [2+2] [CO4]
4. You are assigned to perform sentiment analysis on customer feedback during a product launch. Below are five sample tweets along with their corresponding sentiment labels:

T1: "This product is revolutionary. I absolutely love it!" — Positive

T2: "The item broke after one use. Completely disappointed." — Negative

T3: "Not bad, but could be better. Still worth the purchase." — Neutral

T4: "Great quality and fast shipping. Highly recommend!" — Positive

T5: "Terrible experience. Will not be buying again." — Negative

Use the Naïve Bayes algorithm to classify the sentiment of the following tweet: "The product was good at first, but now it doesn't work."

Describe the process of training the Naïve Bayes model using the given sample tweets and explain how the model would classify the new tweet.

Assume Smoothing: Explain how Laplace smoothing would affect the likelihood calculation for rare words in this context. [4] [CO3, CO4]

---Best of Luck---